

Image properties

1.1 Megapixel

Most digital camera users, brag about their megapixel count, which is also the most misunderstood aspect of digital cameras. A pixel is like a single tile on a mosaic, only that the tiles are so small compared to a mosaic that the resulting image appears very sharp.

In cameras, a pixel is used to denote a single tile on the image sensor that receives the image. A CCD or CMOS sensor in the camera has an array of these tiles that receive the light and convert it into electrical signals that are processed to give the picture. One such array of 2048x1536 will give a pixel count of 31,45,728 pixels, that is, 3.1 megapixels (MP).

However, the number of megapixels in a camera is a very mean measure of the quality of the images that it produces. A higher MP count does not necessarily signify a better image quality, but both the consumer and manufacturers are caught up in a megapixel race. The quality of the lens, algorithms used to make the photo from raw data collected by the sensor, and the focusing mechanism are some of the more important factors contributing to the quality and sharpness of the image – factors where improvements in the image will be much more apparent than cramming in ever more pixels.

For all practical purposes, 3 MP is adequate, while 6 MP is ample – anything higher is moving towards generosity. Another important factor to consider here is how large the sensor itself is. Consumer digital cameras have tiny image sensors – measured in microns, which is a millionth of a metre. Most sensors are less than 10 microns across. This is largely due to the size of the cameras, which are designed at least to fit in your pocket.

Similarly, larger dSLR cameras have much larger sensors, sometimes the size of a conventional frame in a roll of film. Smaller sensors are also cheaper to manufacture. The size of the sensor itself is often overlooked while making decisions of which camera to buy, but it plays a decisive role in the quality of the image.